# Dr. Joseph Nagel

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PyTorch · Lightning · TensorFlow · transformers · scikit-learn · OpenCV · pandas · MLflow · Git · Docker · SQL · AWS · GCP · Claude

## Experience

since 2025 **Staff ML Engineer**, Moody's Analytics, *Munich*
*Geospatial AI for Property Risk Assessment*
- *Finetuned ViT foundation models on aerial imagery with LoRA and other PEFT techniques*
- *Improved the quality of model predictions in the context of international expansion*
- *Ensured that existing models and training pipelines remain state of the art*

2024–2025 **Cofounder / Head of AI**, Validaitor, *Munich*
*Startup for Safe and Trustworthy AI*
- *Helped raising €700k of pre-seed funding*
- *Led a small team of engineers developing a framework for AI trust scores*
- *Implemented a platform for adversarial robustness assessment in the context of tabular data*
- *Finetuned and deployed LLMs for detecting and classifying toxicity of AI-generated texts*

2021–2023 **ML Research Engineer**, Epic Games, *Munich*
*R&D in Generative AI for 3D Animation*
- *Productized the C++ optimization problem solver for animating human avatars in the Unreal Engine*
- *Improved motion capture through VAE-based prior models (90% reduction of end user complaints)*
- *Developed RNN and TCN algorithms for time series forecasting of future animation frames*
- *Enabled the style transfer between different emotions in an animation with GANs and DDPMs*

2018–2021 **Senior AI Scientist**, Airbus, *Munich*
*R&D for Robust Deep Learning and Computer Vision*
- *Initiated and led a research project on safety-critical AI*
- *Realized a confidence estimation POC for surrogate models of the landing distance*
- *Optimized the robustness of a CNN classifier for visual collision avoidance during autonomous aircraft taxi*
- *Developed a 3D detector for material defects in CT scans of additive manufacturing (30% cost reduction)*

## Education

2012–2017 **PhD in Computational Science & Engineering**, ETH Zurich, *Switzerland*
*Research in Machine Learning for Physical Simulations*
**Thesis:** *"Bayesian Techniques for Inverse Uncertainty Quantification"* (*doi:10.3929/ethz-a-010835772*)
**Distinctions:** *ETH Medal, Prize by the Swiss Community for Computational Methods in Applied Sciences*
**Topics:** *Uncertainty Quantification • Statistical Inference • MCMC • Inverse Problems*
**Projects:** *Forecasting of urban river floods for risk assessment and decision-making • Dam breach uncertainty analysis from simulations and historical failure data • Prediction of the structural performance of masonry walls with brick-level data • Damage identification for a civil NASA aircraft from in-flight measurements*

2005–2011 **Diploma in Physics**, University of Bonn, *Germany*
*Specialization in Theoretical Physics and Scientific Computing*
**Thesis:** *"Lattice Quantum Chromodynamics on GPUs"*, **Grade:** *1.4*